FR-07 – Step 1: Identify input & output variables

Spec used

Source: `eshop-sut/README.md` (FR-07, plus the related FR-06, FR-21, FR-23, FR-24) and `api_specification.md` (4.1, 4.2). Black-box: spec documents only, no source code.

> **FR-07: Shopping cart**
>
> - Shows the product list with columns **Product (Sản phẩm)**, **Unit price (Đơn giá)**, **Quantity (Số lượng)** with +/− buttons, **Line total (Thành tiền)**, **Actions (Thao tác)**.
> - Adding the same product again increases its quantity; it does not create a new row.
> - The **Delete** button must show a confirmation dialog first.
> - A **Continue shopping (Tiếp tục mua sắm)** button returns to the home page.
> - The total is labelled exactly **"Tổng cộng"** (not "Tổng tạm tính").
> - An empty cart shows an illustration and a clear message.
>
> **FR-06:** the product detail page has a **Quantity** input that accepts **positive integers only, minimum 1**, and an **Add to cart** button that gives visual feedback (toast or badge update).
>
> **FR-21:** money always uses `₫` with thousand separators; destructive buttons are red; exactly one `<h1>` per page.
>
> **FR-23:** the "Giỏ hàng" navbar link shows a **badge with the number of products in the cart**; breadcrumb required on the Cart page.
>
> **FR-24:** visual feedback after "Add to cart"; confirmation dialog when deleting; empty state with icon/illustration and a friendly message.
>
> **API 4.1 / 4.2:** `GET /api/cart` and `POST /api/cart` require `Authorization: Bearer <token>`. `POST /api/cart` body: `{"id", "name", "price", "quantity"}`.

Defaults applied (change any you disagree with)

1. FR-06's quantity rule (positive integer, min 1) is applied to the cart: quantity in the cart must also stay ≥ 1.
2. Variables from FR-06, FR-21, FR-23 and FR-24 are included only where they concern the cart.
3. UI variables first; API-only variables (price, name and quantity sent by the client, auth token) are included and marked as assumptions.
4. Each distinct message, redirect or display is its own output row; wording is an assumption where the spec gives none.

Input variables

ID	Variable	Condition / constraint	Type	From spec or Assumption
FR-07_IN_01	Product being added	Must be a product that exists in the catalog	Set	Spec

ID	Variable	Condition / constraint	Type	From spec or Assumption
FR-07_IN_02	Quantity entered on the product detail page	Whole number ≥ 1 (no decimals); maximum not specified, probed from 99999999999 upward	Number	Spec (FR-06) + Tester decision (D2, D9)
FR-07_IN_03	Where "Add to cart" is clicked	Set: {product detail page (uses IN_02), home page (quantity 1)}	Set	Spec (detail page) + Assumption (home page button)
FR-07_IN_04	Product already in the cart when added	Set: {not in cart, already in cart}	Set	Spec
FR-07_IN_05	"+" button on a cart row	Clicked n times	Number	Spec
FR-07_IN_06	"−" button on a cart row	Clicked n times; disabled at quantity 1 (rows have a Delete button)	Number	Spec + Tester decision (D3)
FR-07_IN_07	Delete button on a cart row	Set: {clicked, not clicked}	Set	Spec
FR-07_IN_08	Answer to the delete confirmation dialog	Set: {confirm, cancel}	Set	Spec (dialog) + Assumption (cancel option)
FR-07_IN_09	"Continue shopping" button	Set: {clicked, not clicked}	Set	Spec
FR-07_IN_10	Number of distinct products in the cart	Integer ≥ 0	Number	Spec (empty state) + Assumption
FR-07_IN_11	Login state when using the cart	Set: {guest, logged in}; guests must be able to use the cart	Set	Tester decision (D1)
FR-07_IN_12	Page reload / leaving and returning to the cart	Set: {no reload, reload, navigate away and back, log out and in}	Set	Assumption
FR-07_IN_13	`price` sent in `POST /api/cart`	Ignored: the server must use the catalog price	Number	Tester decision (D7); field from API 4.2
FR-07_IN_14	`name` sent in `POST /api/cart`	Ignored: the server must use the catalog name	Text	Tester decision (D7); field from API 4.2
FR-07_IN_15	`quantity` sent in `POST /api/cart`	Whole number ≥ 1	Number	Assumption (FR-06 rule applied to the API)
FR-07_IN_16	`id` sent in `POST /api/cart`	Must be an existing product ID	Number	Assumption (field from API 4.2)
FR-07_IN_17	Auth token on cart API requests	Set: {valid, missing, invalid/expired}	Set	Spec (API 4 header)
FR-07_IN_18	Quantity typed directly into a cart row	Whole number ≥ 1; same maximum as IN_02	Number	Tester decision (D6)

Output variables

ID	Variable	Condition / constraint	Type	From spec or Assumption
FR-07_OUT_01	Cart table columns	Exactly: Sản phẩm, Đơn giá, Số lượng, Thành tiền, Thao tác	Set	Spec

ID	Variable	Condition / constraint	Type	From spec or Assumption
FR-07_OUT_02	Unit price shown on a row	Equals the catalog price; `₫` with thousand separators	Number	Spec (FR-07, FR-21)
FR-07_OUT_03	Quantity shown on a row	Equals the total quantity added for that product, adjusted by +/− or typing	Number	Spec + Tester decision (D6)
FR-07_OUT_04	Line total shown on a row	Unit price × quantity; `₫` with thousand separators	Number	Spec (column) + Assumption (formula)
FR-07_OUT_05	Cart total value	Sum of all line totals; `₫` with thousand separators	Number	Spec + Assumption (formula)
FR-07_OUT_06	Cart total label	Exactly "Tổng cộng"	Text	Spec
FR-07_OUT_07	Number of rows for one product	Exactly 1	Number	Spec
FR-07_OUT_08	Delete confirmation dialog	Set: {shown before deleting, not shown}	Set	Spec (FR-07, FR-24)
FR-07_OUT_09	Row after a delete action	Set: {removed (confirm), kept (cancel)}	Set	Spec + Assumption (cancel)
FR-07_OUT_10	Redirect after "Continue shopping"	Home page	Set	Spec
FR-07_OUT_11	"Continue shopping" button label	Exactly "Tiếp tục mua sắm"	Text	Spec
FR-07_OUT_12	Empty cart state	Illustration/icon + clear message	Set	Spec (FR-07, FR-24)
FR-07_OUT_13	Feedback after "Add to cart"	Toast or badge update shown	Set	Spec (FR-06, FR-24)
FR-07_OUT_14	Cart badge number in navbar	Equals the total number of units in the cart	Number	Spec (FR-23) + Tester decision (D4)
FR-07_OUT_15	Error: invalid quantity	Shown when IN_02 or IN_18 is not a whole number ≥ 1, or is above the maximum	Text	Assumption (wording; rule from FR-06)
FR-07_OUT_16	Cart contents after reload / navigation / re-login	Set: {kept, lost}; expected lost (provisional)	Set	Tester decision (D5, provisional)
FR-07_OUT_17	Response to cart API without a valid token	Rejected (e.g. HTTP 401/403); cart unchanged	Set	Spec (token required) + Assumption (status code)
FR-07_OUT_18	Cart API response to client-supplied data	Price and name: catalog values used, client values ignored. Invalid quantity or id: request rejected	Set	Tester decision (D7) + Assumption
FR-07_OUT_22	"−" button state at quantity 1	Disabled; row not removed	Set	Tester decision (D3)
FR-07_OUT_19	Delete button colour	Red	Set	Spec (FR-21)
FR-07_OUT_20	Breadcrumb on the Cart page	Present	Set	Spec (FR-23)
FR-07_OUT_21	Page heading	Exactly one `<h1>`	Number	Spec (FR-21)

Decision log (spec silent, decided by the tester)

Bugs raised from these decisions must say they rest on a tester decision, not on the spec.

#	Question	Decision
D1	Must the user be logged in to use the cart?	No. **Valid:** a guest can use the cart. **Invalid:** a guest is blocked. (IN_11)
D2	Maximum quantity?	Not specified. Probe the ceiling: start at `99999999999` and increase until the system rejects or breaks. (IN_02, IN_18)
D3	"−" at quantity 1?	Each row has a Delete button (FR-07), so "−" must be **disabled at 1**; it must not remove the row. (New output OUT_22)
D4	What does the badge count?	**Total units** (sum of quantities). (OUT_14)
D5	Should the cart survive reload / re-login?	**Provisional: no** (cart is expected to be lost). To revisit: API 4.1 (`GET /api/cart`) suggests a server-side cart, which would imply it should survive. (OUT_16)
D6	Can quantity be typed directly in the cart?	**Yes**, both typing and +/− must work. (New input IN_18)
D7	Client-supplied price/name in `POST /api/cart`?	Only **backend recalculation** is acceptable: the server must use catalog values. (IN_13, IN_14, OUT_18)
D8	Merge when adding from home page and detail page?	Depends on whether the cart survives page changes (D5). **Decided at execution.**
D9	Quantity edge forms?	Quantity must be a **whole number**; decimals (`1.5`) are invalid. Leading zeros (`007`), signs (`+3`) and scientific notation (`1e2`) are handled as classes in step 2.